

Yuzhe Zhang

Personal Information

Date of birth: January 10, 1998

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Languages: Chinese (native), English (professional working proficiency)

Summary

I am a junior researcher focusing on **axionlike dark matter** detection using **nuclear magnetic resonance** and **quantum sensing** techniques. Within the Cosmic Axion Spin Precession Experiment (CASPER) collaboration, I have contributed to experiments, data analysis, and simulations, resulting in peer-reviewed publications and presentations at international conferences and seminars. Over the years, I have developed expertise in precision measurement, programming, and electronics. I also prioritize collaborative research, clear scientific communication, and teaching.

Education

Doctoral studies in Physics 11.2022–Present

Johannes Gutenberg University Mainz

Thesis: *Search for an Axionlike Dark Matter Candidate Using Nuclear Magnetic Resonance*

Master of Science, Physics 10.2020–10.2022

Johannes Gutenberg University Mainz

Thesis: *Search for Axion-like Particles by Performing NMR on a Thermally Polarized Sample*

Bachelor of Science, Physics 09.2016–06.2020

University of Science and Technology of China

Thesis: *Data Analysis in Earth Rotation Measurements Based on a Nuclear Spin–Electron Spin System*

Research

Cosmic Axion Spin Precession Experiment (CASPER)

10.2020–Present

GSI Helmholtz Institute Mainz & Johannes Gutenberg University Mainz

Advisor: Prof. Dr. Dmitry Budker

Contribution: Experiment, simulation, and data analysis. Design and optimization of axionlike dark matter detection strategy. Cross-team coordination within the CASPER collaboration.

Rb–Xe Comagnetometer Experiment

09.2019–06.2020

University of Science and Technology of China

Advisor: Prof. Dr. Dong Sheng

Contribution: Assistance in the experimental search for spin-gravity coupling. Analysis of co-magnetometer data.

Global Network of Optical Magnetometers for Exotic physics (GNOME)

06.2019–08.2019

Jagiellonian University

Advisor: Prof. Dr. Szymon Pustelny

Contribution: Simulation and data analysis for the GNOME collaboration.

Catalyst Research for Water Splitting

08.2017–06.2019

University of Science and Technology of China

Advisor: Prof. Dr. Jie Zeng

Contribution: Synthesized catalysts, prepared electrodes, and tested water-splitting performance.

Selected Publications

[1] J. Walter, O. Maliaka, Y. Zhang, et al., “Search for axionlike dark matter using liquid-state nuclear magnetic resonance,” *Phys. Rev. D*, vol. 112, p. 052 008, 5 Sep. 2025. DOI: 10.1103/39nc-vr9m

- Commissioning results of the CASPER-Gradient experiment.
- Co-first author (equal contribution; 3 authors). Contributions: apparatus construction and commissioning; data acquisition; analytical calculations and data analysis; manuscript preparation; coordination within the collaboration.

[2] Y. Zhang, D. A. Tumturk, H. Bekker, et al., “Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments,” *Annalen der Physik*, vol. 536, no. 1, p. 2 300 223, 2024

- Theoretical considerations on frequency scanning for axion haloscopes.
- First and corresponding author. Contributions: all theoretical calculations and analysis; full manuscript preparation; coordination within the collaboration.

Presentations

Talks

- 2026 GNOME workshop
- 2023 Academic Seminar of Chinese Students and Scholars in the German–French Region
- 2022 Invited talk at a Seminar at Institute of Theoretical Physics, Chinese Academy of Sciences
- 2021 16th Patras Workshop on Axions, WIMPs and WISPs

Posters

- 2024 The Fundamentals School – summer school in high-energy and gravitational physics
- 2024 DPG Spring Meeting of the Atomic, Molecular, Quantum Optics and Photonics Section (SAMOP)
- 2023 The conference on dark matter, ions, molecules and atoms
- 2022 17th Patras Workshop on Axions, WIMPs and WISPs

Software Development

AxionBloch – [GitHub link]

- Simulation tool for axion-induced spin dynamics using Bloch equations.
- Contributions: all design and implementation.

Academic / Educational Contributions

2026	GNOME workshop Local co-organizer
2026	Educational publication – <i>Some Rules for Good Scientific Presentations</i> contribution: manuscript translation and editing
2025–2026	Teaching assistant in Advanced Physics Laboratory Course - Magnetic Resonance (undergraduate-to-master-level course)
2024	Educational publication – <i>Some rules of good scientific viewgraphs</i> idea initiation; manuscript writing and editing
2022	17th Patras Workshop on Axions, WIMPs and WISPs Local co-organizer

Skills and Experiences

Experiment / Technical

- **Nuclear Magnetic Resonance (NMR):** Apparatus construction, operation, calibration, and data acquisition.
- **Quantum sensing / precision measurements:** Operating Superconducting Quantum Interference Device (SQUID).
- **Cryogenics:** Managing liquid helium and nitrogen for experiments.
- **Hardware automation:** Programming-based control of laboratory equipment.

Programming / Computational

- **Data analysis:** Processing CASPER experimental data with Python; knowledge of particle physics data workflows from workshops.
- **Simulation:** Physics simulations (spin dynamics, signal modeling) using C++ and Python.
- **Numerical equation solving:** Solving Bloch and Schrödinger equations using C++ and Python.
- **Software tools:** Experience with Git / Docker / $\text{\LaTeX}$ / HDF5.
- **High-performance computing (HPC):** Conceptual understanding and practical usage learned through workshops.

Collaboration / Communication

- **International teamwork:** Collaboration with CASPER members. Hosting weekly data analysis meetings with international colleagues.
- **Mentoring and student support:** Assisting interns and students with laboratory integration and research projects; teaching assistant in university experiment course.
- **Scientific communication:** Co-authoring publications, poster presentations, and assisting in talks. Organization of international conference.

Publication List

Peer-reviewed Journals

- [1] D. Aybas, H. Bekker, D. Budker, W. Ji, O. Kim, Y. Kim, D. F. Jackson Kimball, J. Liu, X. Ma, C. P. Salemi, Y. K. Semertzidis, A. O. Sushkov, K. Wei, A. Wickenbrock, and Y. Zhang, “Cavity, lumped circuit, and spin-based detection of axion dark matter: Differences and similarities,” *Universe*, vol. 12, no. 4, 2026, ISSN: 2218-1997. DOI: 10.3390/universe12040106
- [2] J. Walter, O. Maliaka, Y. Zhang, J. W. Blanchard, G. Centers, A. Dogan, M. Engler, N. L. Figueroa, Y. Kim, D. F. J. Kimball, M. Lawson, D. W. Smith, A. O. Sushkov, D. Budker, H. Bekker, and A. Wickenbrock, “Search for axionlike dark matter using liquid-state nuclear magnetic resonance,” *Phys. Rev. D*, vol. 112, p. 052 008, 5 Sep. 2025. DOI: 10.1103/39nc-vr9m
- [3] Y. Zhang, D. A. Tunturk, H. Bekker, D. Budker, D. F. J. Kimball, A. O. Sushkov, and A. Wickenbrock, “Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments,” *Annalen der Physik*, vol. 536, no. 1, p. 2 300 223, 2024
- [4] J. Walter, H. Bekker, J. Blanchard, D. Budker, N. L. Figueroa, A. Wickenbrock, Y. Zhang, and P. Zhou, “Fast shimming algorithm based on bayesian optimization for magnetic resonance based dark matter search,” *Annalen der Physik*, vol. 536, no. 1, p. 2 300 258, 2024. DOI: <https://doi.org/10.1002/andp.202300258>
- [5] Y. Xing, X. Kong, X. Guo, Y. Liu, Q. Li, Y. Zhang, Y. Sheng, X. Yang, Z. Geng, and J. Zeng, “Bi@sn core-shell structure with compressive strain boosts the electroreduction of co(2) into formic acid,” *Advanced science (Weinheim, Baden-Wurttemberg, Germany)*, vol. 7, no. 22, p. 1 902 989, 2020, ISSN: 2198-3844 (Print). DOI: 10.1002/advs.201902989

Software

- [1] Y. Zhang, *Axionbloch*, version 0.2.7, Open-source software repository, GitHub, 2026

Preprints

- [1] Y. Zhang, *Axionbloch: An open-source python package for simulating axion-induced spin dynamics*, 2026. DOI: <https://doi.org/10.48550/arXiv.2606.23749>

In Preparation

- [1] J. Gao, Y. Zhang, J. Walter, and D. Budker, “Search for earth bound axionlike dark matter,” in preparation
- [2] Y. Zhang, J. Walter, O. Maliaka, and D. Budker, “What can casper-gradient do,” in preparation